Supplementary Information (Online Resource 7)

Article title: How Teams Decide on Human Oversight in AI-Powered Systems: A Mixed-Methods Study

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Qualitative evaluation results

A. Qualitative evaluation

The qualitative research included interviews with five experts in human factors engineering. Table 1 outlines the demographic profiles of the experts interviewed.

Participant #	Expertise	Years of experience	Domain and context
E1	Expert and consultant in human-machine interaction	>20	Military
E2	Human-machine interaction designer in systems that include AI components	>10	Safety critical
E3	Expert in Human Factors Engineering with a strong background in systems engineering, human-machine interaction, and autonomous systems.	>20	Autonomous systems, robotics
E4	AI system development expert	>10	Public-sector safety, surveillance
E5	Data science expert	>20	Military, autonomous systems, swarms

Table 1 – Participants demography

During these interviews, we presented the theory and invited the experts to evaluate it and suggest possible improvements. To facilitate richer feedback, we also prepared a set of complementary questions designed to prompt further reflection and critique, including:

1 – How effectively does the initial phase of this framework support capturing real-world operational context and user diversity? What additional contextual elements (e.g., cognitive workload, environmental stressors) should be included?

2 – Does the framework capture all relevant oversight dimensions (e.g., control, transparency, trust, intervention readiness)?

3 – What user-centered design principles should guide the implementation of oversight mechanisms?

4 – What evaluation methods (e.g., simulation, cognitive walkthroughs, usability testing) are most effective for refining oversight mechanisms?

5 – Have you used structured methods (e.g., MCDA, WWW models, workshops) to resolve trade-offs in past projects?

We conducted the deductive thematic analysis [Proudfoot], using Atlas.ti, on the interviews transcripts to identify key themes corresponding to each stage of the framework. The following sections present the main themes extracted from the interviews, organized by framework stages. We marked each theme extracted from the interviews as fully or partially covered in the original framework or as a new theme that needs to be added to the framework.

Legends:

- **Theme** – A key insight or recurring idea derived from expert interviews.
- **Coverage** – Indicates whether and how the theme is currently addressed in the existing framework. F - fully represented, P - partially represented, or N - absent as a distinct element

- **New** – Highlights a novel decision point or activity suggested by the theme that should be incorporated into the framework.
- **Experts** – Reports the identifiers of distinct experts whose available source material supports the theme.
- **Records** – Reports the number of distinct supporting passages identified for the theme in the available interview transcripts or interview summaries. Multiple passages may come from the same expert, so the number of records may exceed the number of experts.

1) Define system context & constraints

Theme	Coverage	New	Experts	Records
User Role Differentiation	Partial	+	E1, E2, E3, E5	4
Domain and Risk Assessment	Full	[redacted]	E1, E3, E4	3
Ethical and Regulatory Constraints	Full	[redacted]	E1, E4, E5	3
Trust and Perception	Partial	+	E2, E4	2
Environmental constraints	Partial	+	E1, E2, E4	3
Domain specificity	Full	[redacted]	E1, E3, E4	3
Risk factors	Full	[redacted]	E1, E4	2
Perceived responsibility and accountability	Partial	+	E3, E4	2
User motivation	None	+	E3	1

The following provides insight into new themes uncovered during the interviews:

Differentiation of User Roles: It is essential to distinguish between types of users, such as service recipients, system operators, and proactive users, as each role entails different oversight requirements. A one-size-fits-all approach to oversight is insufficient; involvement must be tailored to the user's role and context. E1 Transcript extract – "*Each user type may require a different level of involvement and oversight. For example, a system operator or a professional (like a doctor) may need a higher degree of understanding and control compared to a service recipient*."

Trust and Perception: Users' willingness to engage with the system is strongly shaped by psychological factors, including trust, prior experience, and perceived system reliability. These elements directly impact on how much oversight users are comfortable exercising. E2 Transcript extract – "*It doesn't matter how reliable the system actually is, what matters is how reliable the user perceives it to be.*"

Environmental Constraints: The operational setting, such as high-stress or high-tempo environments like battlefields or traffic control centers, can limit the feasibility of active oversight and situational awareness, necessitating adaptive system support. E2 Transcript extract – " *In battlefield or operational situations there are influences that are very difficult to predict in the laboratory*."

Perceived responsibility and Accountability: Effective oversight relies not only on available tools, but also on whether users have or feel a sense of responsibility for system outcomes. When accountability is unclear or diminished, users may become passive and overly reliant on automation, even when intervention is possible. E3 Transcript extract – " *Another challenging aspect is accountability ... when the pilot feels responsible and how to create conditions in which that responsibility is present*."

User Motivation: High levels of intrinsic or extrinsic motivation (e.g., ethical responsibility, mission urgency) can drive users to maintain engagement with the system despite usability challenges. Motivation thus acts as a critical enabler of sustained oversight behavior. E3 Transcript extract – "*What we saw with our military operators was that, regardless of how much we made things difficult for them, they succeeded, because motivation was a factor."*

2) Establish oversight spectrum

Theme	Coverage	New	Experts identifiers	Records
Clarifying levels of automation	Partial	+	E1, E3, E4	4

Clarifying Levels of Automation (LOA): The traditional HITL/HOTL/HOOTL taxonomy provides an overly simplistic view of human roles in complex systems. There is a growing need for more precise definitions and transparent communication of automation levels, enabling users to clearly understand their expected involvement and responsibilities within the system. E3 Transcript extract – " *Intermediate positions could amount to a hundred possibilities. Once you talk about which routines are automated, the main question is what those levels actually are*."

3) Implement dynamic oversight mechanism

Theme	Coverage	New	Experts identifiers	Records
Situational awareness as a prerequisite	Partial	+	E1, E2, E3	4
Explainability aligned with cognitive load	Partial	+	E1, E2	4
Monitoring and Intervention Tools	Full	[redacted]	E1, E2, E4, E5	4
Bidirectional Interaction – feedback by the users	Full	[redacted]	E1, E4	2
Captive vs. voluntary users	None	+	E2, E3	3
Mitigating over-trust \ Automation Fatigue	Partial	+	E1, E2, E3	3
Adaptive interface design	Partial	+	E2, E4	3
Realtime vs. Retrospective decision oversight	None	+	E1, E2, E4	3

Controlled switching between LOA	None	+	E3, E5	2
Implicit explainability	None	+	E5	1

The thematic analysis of expert interviews revealed a strong emphasis on the need for clear, specific, and actionable definitions of human oversight mechanisms, rather than broad or vague categorizations. This stage of the framework was significantly enriched by novel insights, highlighting the diverse ways in which oversight can and should be operationalized in AI-powered systems. The following themes emerged:

Situational awareness as a prerequisite: Effective oversight begins with the user maintaining an understanding of the system's current actions and status, even under high levels of automation. E2 Transcript extract – " *We have to keep the person at some level of awareness so that, if something happens, they can immediately take control.*".

Explainability aligned with cognitive load: Explanations should be context-sensitive and aligned with the user's expertise and cognitive capacity, avoiding both oversimplification and unnecessary complexity. E1 Transcript extract – "*Users must maintain situational awareness relevant to their tasks. Systems should be designed to support the user's mental model, helping them keep track of what the system is doing and why, without overwhelming them with details*".

Captive vs. voluntary users: In environments like military operations, where users are obligated to use the system, oversight can be mandated. In contrast, civilian systems must earn user trust to achieve active oversight. E2 Transcript extract – *" If we turn to people who are not obliged to use it, and the system is very demanding and difficult to handle, they simply will not use it* ".

Mitigating over-trust: Excessive trust in automated systems can lead to disengagement and automation complacency, which is particularly risky in safety-critical settings. Oversight mechanisms should counterbalance this tendency. E3 Transcript extract – " *If you want me to look and pay attention, you have to think about how to make me do it* ".

Adaptive interface design: Interfaces should be configurable, supporting user-specific alerting, control preferences, and layered transparency, offering simple explanations upfront, with optional access to deeper system logic. E2 Transcript extract – " *As part of training or implementation, we can ask which types of notifications the user would or would not want."*

Real-time vs. Retrospective decision oversight: It is crucial to distinguish between decisions that require real-time explanation (e.g., during a mission) and those that can be reviewed post hoc, allowing a balance between operational efficiency and accountability. E2 Transcript extract – " *Some things need to be part of routine operation. Other elements can be placed behind the scenes so that someone can access them calmly, without time constraints*."

Controlled switching between levels of automation: The ability to switch between automation levels must be governed by clear, role-based permissions to prevent misuse while supporting flexibility and contextual needs. E3 Transcript extract – "*The question of who can switch control levels is very important, just the user? someone else? a third party?*".

Implicit explainability: In many situations, users do not rely on explicit, detailed model-level explanations. Rather, their ability to form shared mental models depends on the system behaving consistently and transparently. When automation becomes unpredictable or opaque, users struggle to understand its actions, weakening their trust and oversight. E5 Transcript extract – "*Even when I show the result without algorithm-level explainability, I still think it has great value and gives the person considerable confidence about what is happening.* "

4) Reveal contradictions

Theme	Coverage	New	Experts identifiers	Records
Oversight vs. Performance	Full		E1, E2	2
Transparency vs. Complexity	Partial	+	E1, E2	3
Trust vs. Automation	Partial	+	E1, E3	2
Performance vs. explainability	Full		E5	1
Usability vs. oversight depth	Full		E1, E2	3

Transparency vs. Complexity: Providing more detailed information can improve access to the basis of system behavior while increasing the cognitive demands placed on users. Layered explanations can address this tension by presenting concise information during routine interaction and allowing users to access greater detail when needed. E2 Transcript extract – " *Perhaps we should allow different levels ... and direct users to another place that is more complex if they want to go deeper*."

Trust vs. Automation: Passive monitoring can make it difficult for users to sustain vigilance and situational awareness, potentially weakening their readiness to intervene. E3 explained: "*Being in the loop only passively ... it is very difficult even just to maintain vigilance".*

5) Negotiate trade-offs

Theme	Coverage	New	Experts identifiers	Records
Use of Structured Tools	Full		None	None

Use of Structured Tools is retained from Version 1 as a literature-informed framework component. Direct expert endorsement of the specific methods was not established in the supplied records. Accordingly, this element is not presented as an expert-derived development or as evidence that the effectiveness of these methods was evaluated.

6) Evaluate and iterate

Theme	Coverage	New	Experts identifiers	Records
Usability and Trust as Evaluation Metrics	Full		E1, E2	4
User-informed evaluation of alternatives	Partial	+	E1, E2	3
Objective and subjective UX metrics	Partial	+	E2	2
Iterative usability testing	Partial	+	E1, E2	3
Post-deployment evaluation	Full		E1, E2, E4	3
Need for continual adaptation	Full		E1, E2, E4	4
Graduated trust	Partial	+	E2, E4, E5	3

The thematic analysis of the evaluation and iteration phase revealed a strong need to go beyond general methodological references. Participants emphasized the importance of defining clear, specific, and actionable evaluation tasks that align with system goals and user needs. As a result, this phase of the framework was significantly expanded based on the following insights:

User-informed evaluation of alternatives: Evaluation should include usability tests and structured feedback to assess how much control or explanation users can realistically manage without compromising system performance. This fosters a negotiation between usability and functional complexity. E2 Transcript extract – " *In these tests we also often combine interviews and try to understand why users did what they did and what they felt interfered with using the system*".

Objective and Subjective UX Metrics: Evaluation must incorporate both quantitative (e.g., task success rate, error counts) and qualitative (e.g., perceived ease of use, system usability scores) measures to holistically assess user experience and oversight effectiveness. E2 Transcript extract – " *There are also subjective measures: I ask about satisfaction, how satisfied the person is, and how complex they think the system is."*

Iterative Usability Testing: Continuous experimentation, such as A/B testing, scenario-based trials, or interactive prototyping, is essential to determine which oversight mechanisms and interface designs yield the best outcomes. E2 Transcript extract – " *I examined alternatives; I never examined only one way of conveying information, but different ways and compared them*."

Graduated Trust: Trust in AI systems develop progressively through repeated exposure to consistent, transparent behavior. Early phases should include stronger oversight, with the option to reduce it as the system proves reliable and users gain confidence. E4 Transcript extract – " *Trust must be gradually built, especially in mission-critical environments. The AI's behavior should be predictable and consistent."*